

Francisco ALBUQUERQUE PICADO

Mathematician, Teacher

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Mathematician by training and by heart. Interested in dispelling the bad reputation Mathematics usually gets. Passionate about adequate and effective of communication of Mathematics. Particularly interested in solving problems of all kinds using Mathematics, and its applications to everyday life.

WORK EXPERIENCE

PRESENT DAY September 2022	Teacher of Mathematics, SAINT DOMINIC'S INTERNATIONAL SCHOOL, Cascais > Taught the following subjects : - IB MYP Mathematics and Extended Mathematics - IB DP Mathematics : Analysis and Approaches (SL and HL) > Created and adapted teaching resources to assist the learning of my students and to develop their spirit of discovery. IB MYP MathematicsIB MYP Extended MathematicsIB DP Mathematics AA
PRESENT DAY November 2012	Science Communicator, CIRCO MATEMÁTICO, ASSOCIAÇÃO LUDUS, Lisboa > Planned and presented live shows in order to pin down the inaccessibility of Mathematics > Collaborated with and led traditional board games <i>workshops</i> Recreational MathematicsOutreachTraditional Board GamesPublic Speaking
PRESENT DAY October 2017	Private Tutor, SELF-EMPLOYED, Oeiras > Private lessons to university-level students (Linear Algebra, Mathematical Analysis, Discrete Mathematics and Topology) and high school students (IB DP MAA SL/HL). > Developed original worksheets and other supplementary material. Linear AlgebraMathematical AnalysisTopologyDiscrete Mathematics
February 2022 February 2020	Data Scientist, DIXTOR CONSULTING LDA., Lisboa > Extracted and treated huge volumes of data coming from the banking sector. > Developed and implemented cryptographic models in order to deal with credential managing. PythonSQLMicrosoft ExcelSpyderVisual Studio CodeIDLE$\LaTeX$

EDUCATION AND TRAINING

PRESENT DAY September 2024	Post-Graduate Diploma in Mathematics Teaching, NOVA UNIVERSITY OF LISBON, Caparica, Portugal > Advanced training in mathematics teaching through various means (technologies, manipulatives, real-world applications, games). > Attendance of the specific pedagogical subjects of the full Master's degree program. Relevant subjects : New Technologies in Mathematics Teaching, Pedagogy and Didactics of Mathematics. PedagogyDidacticsPublic SpeakingGeogebra$\LaTeX$
June 2021	Mathematics (MSc.), FACULDADE DE CIÊNCIAS DA UNIVERSIDADE DE LISBOA, Lisboa, Portugal > Dissertation : "Formas Quadráticas e Testes de Primalidade em <i>Disquisitiones Arithmeticae</i>" (18/20) > Advanced training in Algebra and Discrete Mathematics. Specialization in Number Theory. > Programmed a Python library for my dissertation : https://github.com/ghaleon7/FQB_Gauss Abstract AlgebraDiscrete MathematicsCombinatoricsNumber TheoryResearchPython$\LaTeX$
July 2016	Mathematics (BSc.), FACULDADE DE CIÊNCIAS DA UNIVERSIDADE DE LISBOA, Lisboa, Portugal > Basic knowledge in several areas of Mathematics (Algebra, Mathematical Analysis, Probability and Statistics) > Relevant subjects : Introduction to Number Theory, Discrete Mathematics, History of Recreational Mathematics AlgebraLinear AlgebraMathematical AnalysisGeometryStatisticsFinancial MathematicsEconomics$\LaTeX$

July 2016
June 2016

Curso de Educação em Ciência e História Natural, MUHNAC, Lisboa

> Specific training in the field of science dissemination. Organization of lectures and presentations.

Science Dissemination

Public Speaking

Mathematics

LANGUAGES

Portuguese	●	●	●	●	●
English	●	●	●	●	●
Catalan	●	●	●	●	●
Spanish	●	●	●	●	●
French	●	●	●	○	○

TECHNICAL SKILLS

- > Microsoft Office
- > LaTeX text editor
- > Python programming language (NumPy, Pandas)

PUBLICATIONS

- | | |
|------|---|
| 2026 | Albuquerque-Picado, Francisco "An algebraic analysis of the Rubik's Cheese". In Proceedings of the 8th Recreational Mathematics Colloquium : Gathering for Gardner - Europe. Lisboa : Ludus, p. 183-194 |
| 2026 | Malato, Lia AND Albuquerque-Picado, Francisco "A mathematical hangman game". In Proceedings of the 8th Recreational Mathematics Colloquium : Gathering for Gardner - Europe. Lisboa : Ludus, p. 53-60 |
| 2025 | Malato, Lia AND Albuquerque Picado, Francisco. "O jogo da forca no ensino da matemática". Indagatio Didactica, 17(1), 93-110. https://doi.org/10.34624/id.v17i1.40282 |
| 2022 | Albuquerque Picado, Francisco. «Aritmética y barajas de cartas». En : TEMat, 6 (2022), págs. 83-96. ISSN : 2530-9633. URL : https://temat.es/articulo/2022-p83 . |
| 2021 | Picado, Francisco Manuel Albuquerque "Formas Quadráticas e Testes de Primalidade em «Disquisitiones Arithmeticae»". Dissertação de mestrado. Universidade de Lisboa. URL: http://hdl.handle.net/10451/48855 . |
| 2019 | Picado, F. (2019) "The Book of Numbers : Unveiling Sequences". In Recreational Mathematics Colloquium VI. Lisboa : Ludus, p. 139-148 |
| 2017 | Picado, F. "Shuffling a Polygon". In Silva, J.N., ed. lit. Recreational Mathematics Colloquium V : Proceedings of the Recreational Mathematics Colloquium V. Lisboa : Ludus, p. 111-118 |